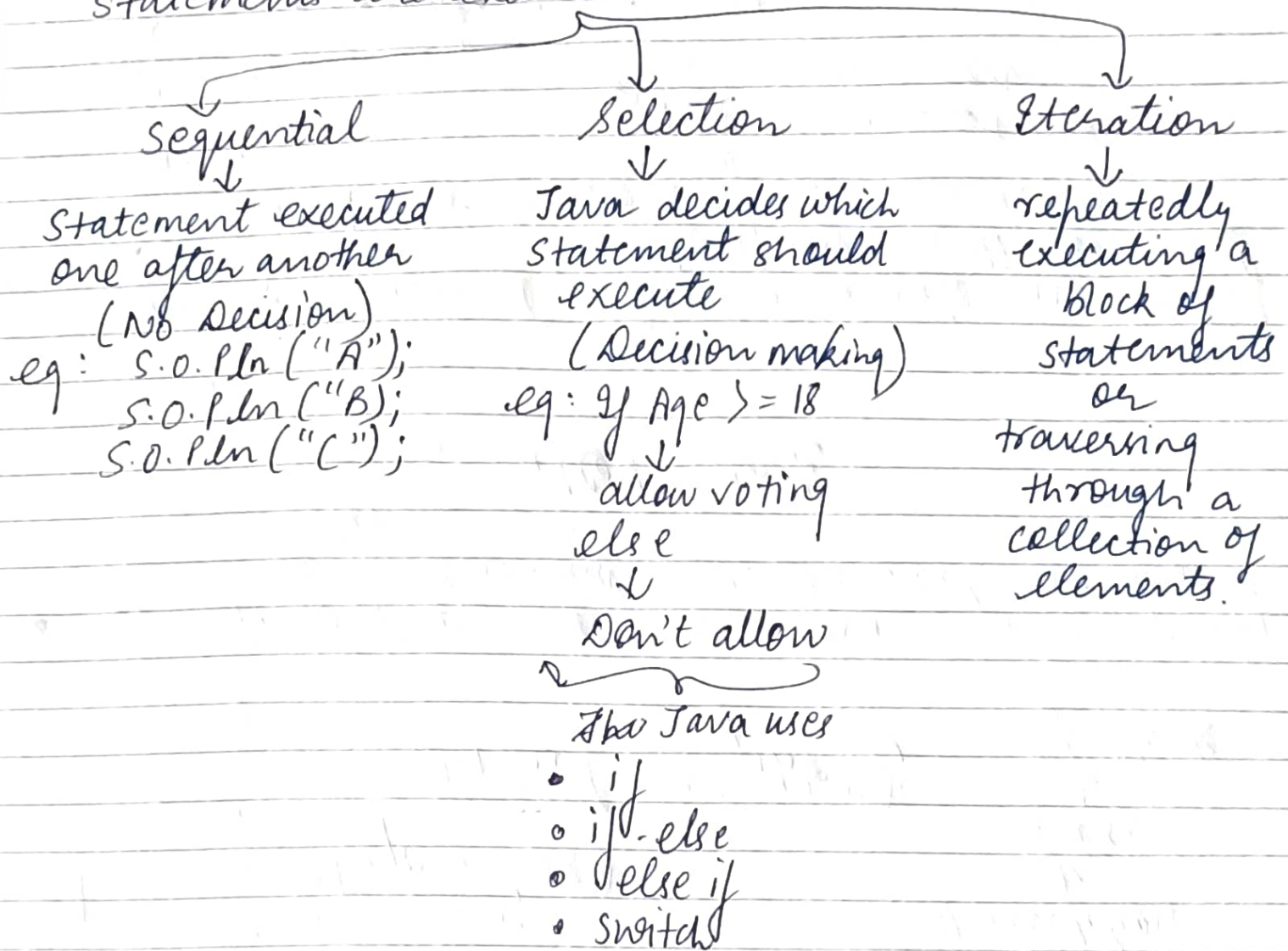


Control Flow Statements

Control Flow determines the order in which program statements are executed.



What is condition?

Condition is something whose answer is True or False.

if Statement

```
if (condition)
{
    statements;
}
```

// If condition is true
↓
execute code

eg:

```
int age = 20;
if (age >= 18)
{
    s.o.pln("Eligible");
}
```

// condition
↓
false
↓
skip block

- If ~~code~~ condition is false, Java skips everything inside & continues with the next statement.

{ } → called block

if-else Statement → used to execute one block of code if the condition is true and another block if the condition is false.

```
if (condition) {
    // Execute the condition is true
} else {
```

only one block execute
never both

// execute the condition is false.

```
}
```

eg:

```
int age = 20;
if (age >= 18)
```

```
{
    s.o.pln("Eligible");
```

```
} else {
```

```
    s.o.pln("Not Eligible");
```

```
}
```

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Nested if statement

A Nested if is an if statement placed inside another if statement.

means:

check the condition only if the first condition is true.

```

if (condition1)
{
    if (condition2)
    {
        statements ;
    }
}
    
```

eg: `int age = 20;`
`int marks = 95;`
`if (age >= 18)`
`{`
 `if (marks >= 90)`
 `{`
 `s.o.pln ("Eligible for Scholarship");`
 `}`
`}`

Nested if with else

eg: if { (age >= 18)

if { (marks >= 90)

{ S.O.P.ln ("Scholarship");

} else

{

S.O.P.ln ("No Scholarship");

}

}

if-else-if Statement

if (condition) {

Statement;

} else if (condition 2) {

Statement;

} else if (condition 3) {

Statement;

} else {

Statement;

}

eg: int marks = 85;

if (marks >= 90)

{ S.O.P.ln ("Grade A");

} else if (marks >= 80)

{ S.O.P.ln ("Grade B");

} else if (marks >= 70)

{

S.O.P.ln ("Grade C");

} else {

S.O.P.ln ("Fail");

}

Switch Statement

The switch statement is used to select one block of code from many possible blocks based on the value of a single variable or expression.

Syntax

```
switch (expression) {  
    case value1:   
        statements;  
        break;   
    case value2:   
        statements;  
        break;   
    case value 3:   
        statements;  
        break;   
    default:   
        statements;  
}
```

int day = 3; compare
switch (day) ^{3 with every case}

Possible value

break; ← stop the switch statement

runs when no case matches (like else) ← default:

eg: int day = 2;
switch (day)
{

case 1 :

S.O.P ln ("Monday");

break;

not break
↓

case 2 :

S.O.P ln ("Tuesday");

break;

fall through
occurs

case 3 :

S.O.P ln ("Wednesday");

break;

default :

S.O.P ln ("Invalid day");

}

Allow datatype

Not allowed

int

char

byte

short

String

enum

float

double

long

boolean